

# Design Of Mechatronics 1 – Work Breakdown Structure

## 1. Planning

### 1.1 Design Sketches

1.1.1 Decide on which mechanisms to use.

1.1.2 Design sketches for each subsystem.

1.1.3 Sketch the integrated subsystems.

### 1.2 WBS

1.2.1 Work Breakdown Structure generation for all project milestones.

### 1.3 Gantt Chart

1.3.1 Plan tasks based on submission times and identify critical tasks.

### 1.4 Bill Of Materials

1.4.1 Identify electronics/mechanical parts and their prices.

## 2. Design

### 2.1 CAD Design

2.1.1 3D Model the first subsystem. (Bi-level sorting)

2.1.2 3D Model the second subsystem. (3D Cartesian Robot)

### 2.2 Manufacturing Plan

2.2.1 Identify the most suitable manufacturing technique/s for each part.

2.2.2 Minimize the use of 3D printed parts & prioritize standard parts.

### 2.3 Actuator Sizing

2.3.1 Identify the actuators in each subsystem.

2.3.2 Perform the suitable calculations based on the loads each actuator is exposed to.

## 2.4 Simulation

2.4.1 Import CAD designs into MATLAB.

2.4.2 Use the toolboxes in MATLAB to run simulations of each subsystem and of the system.

2.4.3 Perform PID control simulations on one of the motors and tune the PID values.

2.4.4 Create a StateFlow model that describes the system.

## 3. Subsystems

### 3.1 Stress Analysis

3.1.1 Perform FEA on critical/load bearing parts of each subsystem.

### 3.2 Manufacturing / Assembly

3.2.1 Manufacture each part using different manufacturing techniques.

3.2.2 Assemble each part correctly as a part of the system/subsystem.

### 3.3 Programming / Electrical Part 1

3.3.1 Programming the microcontrollers for the basic functionality of each subsystem.

3.3.2 Integrate logical and power circuits without damage.

### 3.4 Programming / Electrical Part 2

3.4.1 Implement FreeRTOS.

3.4.2 Establish communication between microcontrollers through ESP-NOW.

## 5. Final Implementation

### 5.1 Integration

- 5.1.1 Align the output of the first subsystem with the input of the second.
- 5.1.2 Establish microcontroller communication between the subsystems.
- 5.1.3 Wire management/Control box implementation.

### 5.2 Testing

- 5.2.1 Run the full system, identify issues, resolve any that come up.